DEFENSE AI INFRASTRUCTURE ENGINEERING

Kenny Sheridan

Infrastructure Product Engineer for AI, robotics, edge, and supercomputing systems.

Systems engineer with hands-on ownership across bare-metal GPU orchestration, edge and datacenter deployments, Kubernetes control planes, RoCE/InfiniBand networking, high-performance storage, reproducible packaging, QEMU/KVM sandboxing, observability, benchmarking, and workload-to-hardware matching. Experience spans medium-sized companies scaling into Fortune 500 operations and pre-seed through post-Series A startups valued from $300M to $1B. Former U.S. Marine Corps meteorology instructor with startup HPC delivery across NVIDIA and AMD platforms for AI/ML, robotics, edge, simulation, and datacenter workloads.

9,000+
GPUs automated across providers and clusters

800G
RoCE and InfiniBand GPU fabric experience

AI
Training and inference workload placement

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EXPERIENCE

Member of Technical Staff - Infrastructure Product | Andromeda

Greater Seattle Area, Remote | March 2026 - Present | Website: andromeda.ai

- Build and scale reproducible high-performance AI infrastructure reliably matched to model training and inference workloads.
- Automate observability and deployment for 9,000+ GPUs across over a dozen globally distributed providers and clusters supporting training and inference capacity for AI/ML, robotics, edge, and datacenter workloads.
- Develop bare-metal GPU orchestration, hardware lifecycle workflows, distributed control planes, and developer-facing platforms for maintainable infrastructure product delivery.
- Use Nix to build reproducible, secure package and deployment environments; use QEMU/KVM for sandboxing, isolation, and repeatable infrastructure validation.
- Create rich product documentation for customer-facing infrastructure workflows, designed for online hosting, mobile viewing, and clear operational adoption.
- Design virtualization and multi-tenant compute isolation patterns that make large-scale GPU infrastructure safer, more repeatable, and commercially usable.
- Drive performance engineering, real-world benchmarking, longitudinal system modeling, caching strategy, monitoring, storage evaluation, and workload placement so the right models land on the right hardware and serve quickly.

Senior Supercomputing Infrastructure Engineer | San Francisco Compute Company

San Francisco, CA | September 2024 - Present | Website: sfcompute.com

- Led automated bring-up for 2,000 NVIDIA H100 GPUs, moving bare metal into operational Kubernetes clusters through a single-command Rust-based deployment workflow.
- Scaled onboarding from 8 nodes to hundreds of GPU nodes within weeks by eliminating manual provisioning steps across hardware, networking, and company infrastructure integration.
- Deploy distributed supercomputing infrastructure globally for GPU marketplace capacity, with emphasis on scalable utilization, reliability, and operational repeatability.
- Optimize compute, network, and high-performance storage layers for large-scale AI and HPC workloads, including real-time troubleshooting and root-cause analysis across distributed systems.
- Build performance testing suites and custom Kubernetes controllers/operators for bare-metal GPU cluster orchestration, resource management, near-metal performance validation, and reproducible systems setup.

Senior AI and HPC Infrastructure Engineer | TensorWave

Las Vegas, NV | May 2024 - July 2024 | Website: tensorwave.com

- Architected on-premises AMD MI300X GPU clusters using EPYC CPUs and RDMA over Converged Ethernet on traditional TCP/IP networks.
- Benchmarked NVIDIA InfiniBand and AMD RoCE designs, including high-bandwidth all_reduce testing over 800G switching infrastructure.
- Designed vendor-agnostic AI/ML infrastructure patterns capable of scaling toward hundreds of nodes while reducing accelerator lock-in.
- Created reproducible deployment documentation for high-performance GPU cluster setup, configuration, and operational handoff.

Senior Hardware Infrastructure Automation Engineer | ServiceNow

SKILLS

Linux · Rust · Nix · Secure packaging · QEMU · KVM · Sandboxing · Kubernetes · Operators · GitOps · Argo CD · Bare metal · Hardware lifecycle · Redfish · NVIDIA H100 · AMD MI300X · CUDA · ROCm · NCCL · RoCE · InfiniBand · SONiC · SDN · Tailscale · Netmaker · DDIL-aware edge · NVMe · Weka · VAST · Ceph · Virtualization · Multi-tenant isolation · Model caching · OpenTelemetry · Prometheus · Grafana · FIO · iperf3 · stress-ng · MPI · PyTorch · Simulation infra · Robotics infra

STRENGTHS

Defense AI infrastructure
Build reproducible platforms for AI/ML, robotics, simulation, edge, and datacenter workloads.

Reproducible systems
Use Nix, QEMU/KVM, GitOps, and documentation to make infrastructure repeatable and auditable.

GPU and edge orchestration
Automate bare-metal NVIDIA and AMD fleets across providers, clusters, and constrained environments.

Fabric, storage, and telemetry
RoCE, InfiniBand, SDN, Weka, VAST, Ceph, OpenTelemetry, Prometheus, and Grafana.

Validation-first delivery
Benchmark, stress, profile, model, and match workloads to hardware with

Kirkland, WA | February 2022 - December 2023 | Website: servicenow.com

- Engineered HPC system testing software for distributed enterprise infrastructure, including stress validation, benchmarking, and reliability assessment workflows.
- Led migration of internal automation from Python and Bash to Go, improving efficiency across heterogeneous hardware environments.
- Built Redfish-based hardware SKU auditing and NIC performance tooling for asset validation, 25G/50G/100G/200G Ethernet testing, and backend infrastructure assurance.
- Managed GitLab CI/CD pipelines for cross-department hardware-software co-design, validation, and verification.

Senior Cloud Hardware Performance Test Engineer | ServiceNow

Kirkland, WA | May 2017 - February 2022 | Website: servicenow.com

- Led cloud hardware performance testing across storage, networking, BIOS, firmware, PCIe devices, FPGAs, NVMe, SAS/SATA, Linux filesystems, and scale-out storage platforms including Weka, VAST, and Ceph.
- Collaborated with ODMs, system engineers, CTO stakeholders, product teams, and cross-functional groups to refine infrastructure component roadmaps and specifications.
- Trained three junior engineers and a mid-level manager while improving onboarding, agile practices, and test methodology consistency.

System Administrator | NexLevel Information Technology

Sacramento, CA | August 2015 - May 2017 | Website: nexlevelit.com

- Provided Tier 3 Unix and Windows server support for biometric capture and analysis systems serving 300+ remote clients.
- Recovered critical state records from a compromised storage array, preserving SLA compliance and avoiding substantial penalties.
- Built monitoring scripts and baseline configuration improvements to reduce downtime and stabilize production infrastructure.

Technical Instructor of Meteorology | U.S. Marine Corps

Quantico, VA | May 2007 - June 2015 | Website: marines.mil

- Administered two modular data centers and maintained METMF(R) computing infrastructure supporting mission-critical weather forecasting operations.
- Virtualized Windows instructional environments with VMware, managed WAN-connected remote sensing sites, and deployed Solaris-based mobile weather radar units.
- Produced 300+ surface weather observations, 100+ 96-hour forecasts, and 50+ weather warnings cited in Navy and Marine Corps Achievement Medal recognition.

SELECTED ENGINEERING WORK

Reproducible AI infrastructure environments

Nix | QEMU/KVM | Secure packaging | Model-serving cache paths

Set up reproducible infrastructure environments that use Nix for secure packages, QEMU/KVM for sandboxed validation, and caching strategies to serve AI models quickly.

Multi-cluster HPC observability platform

Rust | Prometheus | Grafana | HPC workloads | Infrastructure monitoring

Custom-built single-pane-of-glass observability system for multi-cluster HPC workloads and infrastructure, designed to integrate with Prometheus and Grafana workflows.

Automated GPU bring-up and onboarding

Rust | Kubernetes | Bare metal | NVIDIA H100

Single-command workflow that deploys hardware, joins company infrastructure, configures networking, and removes manual intervention from large-scale GPU node onboarding.

Multi-node GPU cluster networking

AMD MI300X | NVIDIA | RoCE | InfiniBand | 800G fabrics

Designed vendor-agnostic topologies across AMD and NVIDIA accelerators, including RoCE on TCP/IP networks and InfiniBand benchmarking for large-scale AI/HPC clusters.

LLM and HPC benchmarking CLI

Rust | MPI | PyTorch | DeepSeek-R1 | NVIDIA and AMD GPUs

Resource-efficient benchmarking application for rapid system characterization across InfiniBand, LLM workloads, and heterogeneous GPU infrastructure.

hardware_report

Rust | San Francisco Compute Company | Hardware, orchestration, virtualization

Produces serialized hardware reports for physical infrastructure automation, supporting host introspection, inventory, scheduling inputs, and orchestration workflows.

rocm_gpu_tradecraft

ROCm | GPU operations and command tradecraft

operational evidence.

OPEN SOURCE

Forgejo portfolio

Self-hosted forge for systems architecture and private/public engineering work.

GitHub profile

Public Rust, Nix, GPU, MCP, and developer tooling repositories.

Neovim tooling

Lua-based personal and professional editor configuration used since 2015.

EDUCATION

Air University / Community College of the Air Force

Keesler AFB | 62 credits in meteorology, forecasting, and atmospheric physics

CERTIFICATIONS

Introduction to Blockchain

Amazon Web Services, 2021

Basic Instructor Course

Community College of the Air Force, 2012

Meteorology and Oceanography Analyst Forecaster

Community College of the Air Force, 2008

RECOGNITION

Navy and Marine Corps Achievement Medal

Good Conduct Medal

Letter of Appreciation, Director of Meteorology

Operational reference for engineers working with AMD ROCm tooling, GPU validation, and accelerated compute environments.